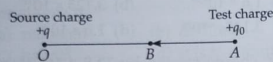


the repulsive force exerted by the charge $+q$. The potential difference between points A and B is

$$V = V_B - V_A = \frac{W_{AB}}{q_0} = \frac{\text{Work}}{\text{Charge}}$$



- 5 J of work is done in moving a positive charge of 0.5 C between two points. The potential difference between these two points is
(a) 25 V (b) 2.5 V
(c) 10 V (d) zero
- The SI unit of electric potential is
(a) ampere (b) volt
(c) joule (d) coulomb.
- A charge of 2 coulomb moves between two points maintained at a potential difference of 1 volt. The energy acquired by the charge is
(a) 0.1 J (b) 0.2 J
(c) 1 J (d) 2 J
- How much is the electric potential of a charge at infinity?
(a) Zero (b) Infinite
(c) One (d) Can't determine

CHAPTER 02

Heat Transfer in Nature

Learning Outcomes... *note*

- Heat and Temperature
- Thermometers
- Transfer of Heat
- Water Cycle

Introduction

We often use the word 'hot' and 'cold' while describing things around us. Suppose you have a bowl of hot soup or a cup of coffee that is too hot. What can you do to cool it? You would probably add something cooler such as ice.

When we mix ice to hot coffee it becomes cooler and ice starts melting. Something is getting transferred between the ice and hot coffee. **This something is actually a kind of energy which transfers from a hot body to a cold body and is known as heat.** This chapter is an attempt to elaborate different aspects of heat, temperature, their measurement and their effect.

Heat and Temperature

Heat is a form of energy which flows from a body at a higher temperature to lower temperature. Temperature is a physical quantity that measures the degree of hotness or coldness of a body. Heat can increase or decrease the temperature. The direction of flow of heat between two bodies is determined by their temperature.

DO YOU KNOW?

Chemical energy stored in fuels or food can be changed into heat energy. For example, burning wood, coal or candle releases heat and even the food we eat gives energy that our body partly use as heat.

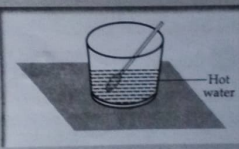
ACTIVITY CORNER



To understand how heat flows from a body at a higher temperature to one at a lower temperature.

- Take some hot water in a tumbler. Dip a metal spoon in this water for some time. How long does it take for the spoon to become hot for you to feel uncomfortable to hold it anymore?

- This shows that heat is transferred from the hot water in the tumbler to the spoon and then to your hand.
- Leave the tumbler with water in it for 10-15 minutes. Feel the water now. Has it cooled down? Why?
- Water has lost its heat to the surroundings. This shows that heat has been transferred from the water to the surroundings.



Measuring Temperature

The process of measuring temperature is called thermometry. Temperature is measured by a device called **thermometer**.

Temperature Scale

There are three scales of temperature used to measure temperature.

- (i) Celsius ($^{\circ}\text{C}$) (ii) Fahrenheit ($^{\circ}\text{F}$) (iii) Kelvin (K)

To measure temperature, generally Celsius scale and Fahrenheit scale are used.

A scale of temperature is constructed on the basis of two fixed temperatures called fixed points. The lower fixed point of the Celsius and Fahrenheit scales is known as the ice point. It is the temperature at which ice melts.

The upper point of Celsius and Fahrenheit scales is known as steam point. It has been given the value of 100° on the Celsius scale and 212° on the Fahrenheit scale.

DO YOU KNOW?

For a given substance, the freezing point of its liquid state and melting point of its solid state has same numerical value.

Kelvin scale is used a lot in astronomy. Temperature in space are often much colder or much hotter than the Earth. The Kelvin scale is good to use for both. On the Kelvin scale, the value of the ice point is 273 K and that of the steam point is 373 K .

Table : The upper and the lower fixed points on different temperature scales.

	Celsius Scale	Fahrenheit Scale	Kelvin Scale
Lower fixed point	0°C	32°F	$\sim 273\text{ K}$
Upper fixed point	100°C	212°F	$\sim 373\text{ K}$

To convert the temperature from one scale to another, the following relation is used:

$$\frac{\text{Temperature on one scale} - \text{Lower fixed point}}{\text{Upper fixed point} - \text{Lower fixed point}} = \frac{\text{Temperature on other scale} - \text{Lower fixed point}}{\text{Upper fixed point} - \text{Lower fixed point}}$$

∴ The relation between Celsius scale, Fahrenheit scale and Kelvin scale are given as :

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} = \frac{T_K - 273}{373 - 273}$$

The relationships between the three different temperature scales (Celsius, Fahrenheit and Kelvin) are also given below.

$$(i) T_F = 9/5 \times T_C + 32 \quad (ii) T_C = 5/9 (T_F - 32) \quad (iii) T_K = T_C + 273 = 5/9 (T_F - 32) + 273$$

ILLUSTRATIONS

- 1 Convert 95°F to degree Celsius.

$$\begin{aligned} \text{Ans.: } T(^{\circ}\text{C}) &= \frac{5}{9} \times (T(^{\circ}\text{F}) - 32) \\ &= \frac{5}{9} \times (95 - 32) = \frac{5}{9} \times 63 = 35^{\circ}\text{C} \end{aligned}$$

- 2 What is the minimum possible temperature? Is there also a maximum possible temperature?

Ans.: The minimum possible temperature is 0 K . There is no limit to maximum temperature.

Thermometers

The most common type of thermometer is 'mercury-in-glass' or **laboratory thermometer**. Such thermometers have a long glass stem with a thin, uniform bore, closed at one end. The other end is attached to a small glass bulb filled with mercury. When the bulb is heated, the mercury expands and flows into the bore of the stem, and looks like a shining thread. The temperature scale is marked on the stem. The thermometer reading is the temperature coinciding with the end of the thread.

The **clinical thermometer**, shown in figure is used to measure body temperature. The temperature of human body normally does not go below 35°C or above 42°C . This is the reason, the thermometer has the range 35°C to 42°C . For other purposes, we use the laboratory thermometers. The range of these thermometers is usually from -10°C to 110°C . The bore of a clinical thermometer has a kink that does not allow the mercury to move back as soon as the temperature changes. The thermometer has to be given a couple of sharp jerks to move the mercury back.

Mercury is commonly used in thermometers because it has several unique properties.

- Its expansion with change in temperature is large and uniform. A large expansion makes it easy to read temperatures on the thermometer. And a uniform rate of expansion makes it easier to graduate (mark the scale) the thermometer.
- It does not stick to glass, so it moves smoothly inside the glass stem.
- It is opaque and shiny, so it is easily visible.
- It remains in the liquid state over a large range of temperature.

Solids and gases are also used in special thermometers needed to measure very high temperatures or in other special circumstances.

Digital Thermometer

Nowadays digital thermometers are widely used which do not contain mercury (As mercury is toxic in nature). A digital thermometer has a plastic body and is easy to use. It provides temperature readings in decimal points for an accurate and specific reading.

It is faster than the normal thermometer in calculating readings.

Transfer of Heat

To carry heat energy from one part of an object to its other part or from one object to another object is called transfer of heat. For example, put a hot object in contact with a cold object. After some time, you would notice that the cold object becomes hot while the hot object becomes less hot. This happens because the hot object transfers some of its heat to the colder object. In fact in all cases heat flows from a hotter object to a colder object.

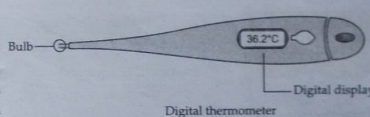
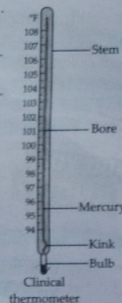
- How does heat flow?

The mechanisms by which heat is transferred from one region to another are called modes of heat transfer. The three modes of heat transfer are as follows:

- (i) Conduction (ii) Convection (iii) Radiation

Conduction

The process of the transfer of heat energy from the hotter end to the colder end of an object without any movement of the object itself is called conduction of heat.



In solids, generally the heat is transferred by the process of conduction of heat.

Matter is made up of molecules. In solids, molecules are tightly packed. They are not free to move, but they can vibrate about their fixed positions. When one part of a solid is heated, the molecules in that part have more energy and vibrate faster. When vibrating more vigorously, they push and pull the molecules of the colder part and make their vibrations more energetic. Thus, energy flows from the hotter to the colder part of the body.

DO YOU KNOW?

Conduction is usually a slow process and can either occur within a body or between two bodies when they are in contact with each other.

Conductors and Insulators

in your own words.

- The materials which allow heat to pass through them easily are good conductors of heat. For example, aluminium, iron and copper.
- The materials which do not allow heat to pass through them easily are poor conductors of heat such as plastic and wood. Poor conductors are also known as insulators.

Both good and bad conductors are used in cooking utensils. The bottom is made of copper or some other metal. Metals are good conductors, so they conduct heat readily from the fire to the material inside the utensil. The handle, on the other hand, is made of bakelite, which is a bad conductor. Therefore, it does not become too hot even when the utensil is kept over the fire.

Similarly materials such as wool and cotton are used to make winter clothing, blankets and quilts. This is because they are made of very thin fibres which have a lot of air spaces. Air is a good insulator, so materials with air spaces are good insulators. That is why traditional quilt makers beat cotton to make it fluffy creating more air spaces and improving insulation.

ACTIVITY CORNER



Experimental investigation

- Take a steel spoon, a geometry box compass, a plastic scale, a pencil and a divider. Dip one end of these articles in hot water and wait for a few minutes.
- Touch the other end and find out which of given articles are conductors and which are insulators. Note down your observation in the following table.

Materials taken	Conductor or Insulator
Steel spoon	
Geometry box compass	
Plastic scale	
Pencil	
Divider	

ILLUSTRATIONS

- 3 At what common temperature, would a block of wood and a block of metal feel equally cold or equally hot when touched?

Ans.: When the temperature of the object is equal to that of the human body, no transfer

of heat takes place between the body and the object. Thus, when the temperature of the block of wood or block of metal is equal to 37°C, i.e., the temperature of the human body, these appear equally hot or cold on touching on account of non-transfer of heat.

- 4 Why an ice box is constructed with a double wall?

Ans.: An ice-box is made of double wall and the space between the walls is filled with some non-conducting material to provide heat insulation, so that the loss of heat is minimized.

- 5 Pieces of copper and glass are heated to the same temperature. Why does the piece of copper feel hotter on touching?

Ans.: We know that copper is a far better conductor of heat than glass. When we touch the hot copper piece, it transmits heat readily

to the hand. But this is not in the case when the hot glass piece is touched.

- 6 Place a safety pin on a sheet of white paper. Hold the sheet over a burning candle until the paper becomes yellow and chars. Now shake off the pin. It will leave a white trace on the paper. Explain this phenomenon.

Ans.: This is due to the reason that the pin, being a good conductor of heat, takes heat from the paper under it and gives away to the surroundings. As a result of this, the paper under the pin is comparatively colder than its remaining part.

Convection

In fluids (liquids and gases together), the molecules themselves carry energy from a hotter region to a colder region. This process of heat transfer is called convection. Convection plays a major role in weather phenomena. For example, the almost direct rays of the sun heat the oceans rapidly. The air near the ocean gets heated and rises. The rising air cools, and the water vapour present in it forms clouds. Cooler air from above sinks, gets heated and rises again to form more clouds, which brings rain.

Heat transfer in fluids occurs almost exclusively through convection. This is because almost all liquids and gases are poor conductors of heat.

ACTIVITY CORNER

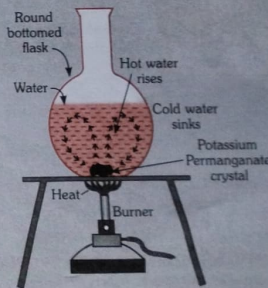


To show convection currents during heating of water.

- We can see the path of convection currents of hot and cold water taking place during the heating of water by dropping a crystal of a coloured substance called

'potassium permanganate' into the water to colour it. This can be done as follows.

- We take a round-bottomed flask and fill it half with water. A small crystal of potassium permanganate is dropped carefully at the bottom of the flask containing water. This potassium permanganate crystal dissolves slowly and forms purple coloured solution around itself. Heat the water at the bottom of the flask by keeping a burner below it and observe the movement of this hot water (which has been coloured purple by dissolving potassium permanganate crystal). We will see the purple streaks of hot water rise from the bottom of flask up to the surface of water and then sink downward near the walls of the flask. These purple coloured streaks seen in the water of flask (which is being heated from below) show the convection current taking place in the water of flask which transfer heat from the bottom to the top.



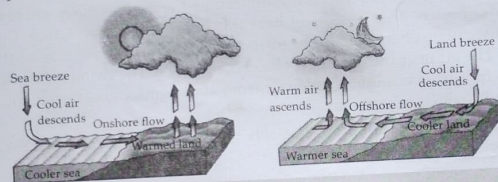
Activity to show the convection currents taking place during the heating of water which transfer heat from bottom to the top.

Applications of Convection

- Ventilators are located near the ceilings of rooms. This allows hot air of the room which has risen up to move out of the ventilator. The fresh cool air from the window rushes in to take the place of rising warm air.
- Air conditioners are installed near the roof. The cool air moves down and the warm air near the surface of the room rises up thereby setting up convectional currents which cool the whole room effectively.

Land Breeze and Sea Breeze

- Convection plays an important role in setting up winds. When some region gets heated, the air over it becomes lighter and moves up. Cold air from neighbouring regions move in to take its place, causing winds. In coastal areas, when the land and the sea absorb heat from the sun during the day, the land heats up faster than the sea. The air above the land becomes hot and moves up, and cooler air over the sea moves in to take its place. This is called a **sea breeze**. At night, the air above the land cools faster than the air above the sea, and air moves out from the land towards the sea. This is called a **land breeze**. These breezes keep the climate moderate in coastal areas.



ACTIVITY CORNER

To show the transfer of heat in air by convection.

- » Fix a lighted candle on a table. Keep one hand at a safe distance above the flame of the candle and the other hand on the side of the flame. You will observe that the hand kept above the candle flame feels quite hot but the hand kept on the side of the flame does not feel so hot. Why does it happen?

The air just above the candle flame gets heated first. This hot air (being lighter in mass) rises upward carrying the heat along with it and the cold air from above (being denser) sinks downward to take the place of hot rising air. This cold air also gets heated by the flame and rises upwards. And more cold air sinks downwards to take its place. This process of hot air rising up and cold air moving down is repeated continuously. By this process, convection currents are set up in the air above the candle flame which carry more and more heat upwards. This transfer of heat by moving air makes our hand kept above the candle flame feel very hot.

- » The most interesting part is that, air can transfer the heat from a source by convection neither on the sides nor in the downward direction (below the candle). This means that no convection currents of hot air take place on the sides of candle flame due to which the hand kept on the side of the flame does not feel so hot. However small heat is felt by the hand kept on the side of the candle flame due to the transfer of heat of candle flame by the process of radiation.



DO YOU KNOW?

- » Mercury, though a liquid is heated by conduction.
- » A hot bar placed under a running water loses heat by conduction between the surface of bar and water and not by convection within the water.

Radiation

When you sit in front of a fire, you will feel warmth of the fire. However, the air between you and the fire is a bad conductor of heat and does not get heated up much. Therefore, heat is not reaching you by conduction. It also cannot reach you by convection since convection currents rise upwards.

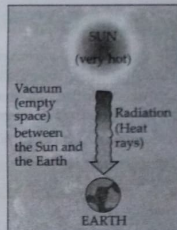
Therefore there is another method by which heat energy reaches you. This method is called radiation of heat.

Radiation is the only process of heat transfer that can occur in the absence of any form of medium. It refers to the transmission of energy as electromagnetic radiation. Radiation is the fastest mode of transfer of heat.

All hot objects—solid, liquid or gas emit radiant energy. When radiant energy falls on another object, it is partially absorbed and partially reflected. The amount of heat absorbed or radiated only depends upon the nature and the colour of the object. Dark colours are good absorbers of radiant heat, while light colours are poor absorbers and good reflectors of heat.

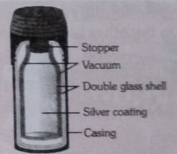
Therefore every hot object emits (gives out) invisible heat rays in all directions. So, heat can be transferred from a hot body to a cold body by radiation even if there is no material medium (like solid, liquid or gas) between them.

Heat from the Sun reaches the Earth by the process of radiation. In this case, conduction or convection is not possible because of the vacuum between the Sun and the Earth.



DO YOU KNOW?

- » The thermos flask minimizes transfer of heat by conduction, convection and radiation. The silver coating on the inner surface of bottle prevents heat transfer by radiation, the vacuum between its double wall prevents heat transfer by convection and the thinness of the glass wall stops heat leaving and entering the flask by conduction.



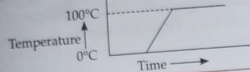
Change of State

- » Matter exists in three states viz, solid, liquid or gas. The physical state of matter can be changed by adding or removing heat energy from it. For example, when heat is added to ice, which is in solid state, it changes to water which is in liquid state. By adding heat further, water can be

COMPETITION WINDOW

vapourised so as to change it into gaseous state. Similarly, a substance in gaseous state can be brought into liquid and then solid state by successively removing heat energy from it. When temperature of melting ice is recorded, say after every 10 minutes and

a graph of temperature versus time is plotted, we get a graph as shown in figure.



► Note that the temperature remains constant at 0°C until all the ice melts. After that the temperature of water increases and remains constant again at 100 °C till all the water vaporizes.

Kinds of Clothes We Wear in Summer and Winter

We prefer dark coloured clothes in winter because dark colours absorb heat and retain it for some time, whereas light colours reflect back most of the heat that falls on it. Thus, dark coloured clothes give us more comfort in winter, while we feel comfortable wearing light coloured clothes in summer.

The colour of a body is also responsible for the amount of heat it absorbs.

ILLUSTRATIONS

7 If air is a bad conductor of heat, why do we not feel warm without clothes?

Ans.: Air is a poor conductor of heat. Moving air carries heat due to convection. Thus, when we are without clothes, the moving air in contact with our bodies carries away the heat from our bodies and hence we feel cold.

8 Why do we prefer white or light coloured clothes in summer than dark coloured clothes?

Ans.: White or light coloured clothes absorb the least heat from the Sun and hence keep us comfortable in summer. On the other hand, dark coloured clothes absorb more heat from the Sun and keeps us hot in summer. Thus we prefer white or light coloured clothes in summer more than dark coloured clothes.

Matter Expands on Heating and Contracts on Cooling

Matter expand when heated. When a solid is heated, its particles gain energy and begin to move faster and distance between them increases and they expand. Gas expand more compared to solid and liquid.

On the other hand, matter contracts on cooling, causing decrease in volume or size of the matter.

DO YOU KNOW?

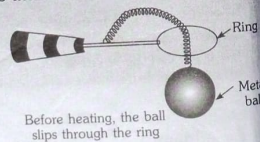
When the same amount of heat is given to different substances, they expand by different amounts. Gases (like water vapour, CO_2 , O_2 etc) expand the most, liquids (like water, Hydrogen peroxide, etc.) expand less and solids (like iron, steel etc.) expand the least.

ACTIVITY CORNER

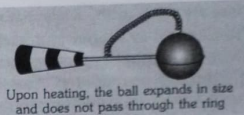
Now heat the ball with spirit lamp for some time and then keep the ball again on the ring. The ball does not pass through the ring.

Gravesande's ring and ball experiment

► In this apparatus of a ring and ball, the internal diameter of the ring and the external diameter of the ball are equal. When the ball is kept on the ring it just slips through the ring as shown in figure.

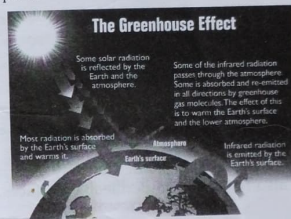


► From this experiment, it is clear that the solids expand on heating. Now, allow the ball to cool to room temperature and place it again on the ring, then the ball passes through the ring easily. This shows that solids contract on cooling.



Greenhouse Effect

► Solar radiation reaches the Earth's atmosphere and some of this is reflected back into space.



► The part of the Sun's energy absorbed by the land and the ocean, causes the Earth's

COMPETITION WINDOW

surface and atmosphere to heat up.

- Heat radiates from the Earth's surface towards space.
- Some of this heat is trapped by greenhouse gases (CO_2 , NO_2 , etc.) in the atmosphere making the Earth warm enough to sustain life.
- Human activities such as burning fossil fuels, agriculture by products, deforestation are increasing the amount of greenhouse gases released into the atmosphere.
- This trapping of extra heat, which causes the Earth's temperature to rise is known as Greenhouse effect.

Water Cycle

We already know that water exists in three forms in nature. As a liquid, water fills the oceans, rivers, lakes, ponds, streams on the Earth and it also exists in the soil and underground. As a solid, it forms snow, ice sheets, and glaciers in the mountains and the polar regions. As a gas, it exists in the form of water vapour in the Earth's atmosphere.

During the summer days, when the average daily air temperature rise above the freezing point, some of the snow and the ice begin to melt and are converted into water. Thinning of the ice layer is caused by heat transfer and by melting at the top or bottom surface (or both). This is due to the Sun's radiation and the water flows down into the water streams and rivers, and ultimately into the oceans. The melted ice is replenished by fresh snow during winter.

There are many ways that water moves between pools — liquid water evaporates into water vapour, is transported through the atmosphere, condenses to form clouds and returns to the surface as precipitation like rain or snow. Water can flow across the land, soak into the ground, then be absorbed by plants, and return to the atmosphere. Water can even return to the atmosphere from ice or snow, without becoming liquid.

Here are some of the ways that water moves, or fluxes, and the terms we use to describe them:
From liquid to vapour and back again.

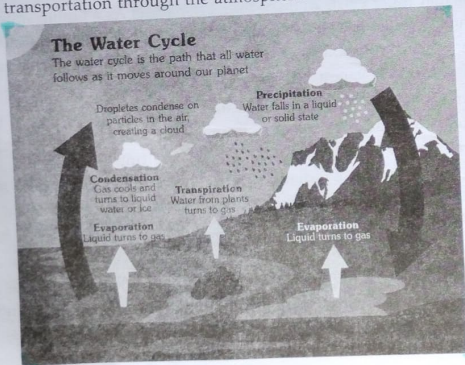
- **Evaporation:** Liquid water can turn into water vapour, like moisture in the atmosphere, through evaporation. Any liquid water can evaporate — from falling rain to water in the ocean or in a puddle on the sidewalk. Evaporation happens when water absorbs energy from the sun, the atmosphere, or even our own body heat.
- **Transpiration:** Liquid water evaporates from the leaves of plants in a process called transpiration. Much of the water that plants take up through their roots and store eventually leaves the plant this way.
- **Condensation:** Water vapour can condense into liquid. Water vapour condenses into water droplets to form clouds or to form dew on the ground. Condensation happens when there is more water vapour than the air can hold at the current temperature.

From vapour to solid and back again.

- **Sublimation:** Snow or ice can turn directly into water vapour — without melting — through sublimation.
- **Deposition:** Water vapour can turn directly into ice without condensing first. This is called deposition and can result in phenomena like hoar frost.

Flowing and falling: Moving from place to place.

- **Precipitation:** Once water has condensed in the atmosphere, it eventually falls back to the Earth through rain, snow, or other types of precipitation.
- **Runoff and streamflow:** Water can flow over the surface of the land. This may be through lakes, streams, or as ice in glaciers. Water can also flow over impermeable surfaces, like concrete.
- **Infiltration:** Water on the ground surface can move into the soil.
- **Percolation:** Water that has infiltrated the soil can continue moving downwards into the groundwater.
- **Transportation:** Water can be transported through the atmosphere as water, ice, or vapour. Clouds moving across the sky are a visual example. Atmospheric rivers are a type of concentrated transportation through the atmosphere.



The water cycle describes the continuous movement of water in the Earth and atmosphere. Water moves naturally in a water cycle, a cycle that involves the continuous circulation of water in the Earth-atmosphere system. Although the total amount of water within the cycle remains essentially constant, its distribution among the various processes is continually changing. Thus, the water cycle helps in redistributing and replenishing water in river, lakes and ocean. Rainwater that falls on the surface of the Earth flows into ponds, lakes, rivers, and oceans or seeps into the ground.

Evaporation

- » The energy required for evaporation is taken from the surroundings.
- » Evaporation of liquid takes place at all temperatures but the rate of evaporation is different at different temperatures. The rate of evaporation increases with increase in temperature.
- » When some liquid evaporates, the average speed of the remaining molecules will decrease because the more energetic

COMPETITION WINDOW

molecules have left and temperature of the liquid dropped.

- » When a liquid boils, vapours are formed. The boiling of liquid takes place at a definite temperature called boiling point. At boiling point, a substance in liquid state changes to gaseous state.
- » In both evaporation and boiling, a liquid changes to gaseous state yet both the process are different.

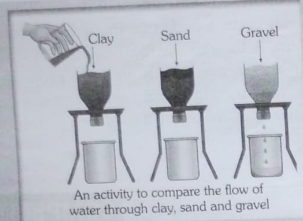
Seepage of Water Beneath the Earth

ACTIVITY CORNER



Investigating water seepage through different soil types.

- » **Prepare the bottles:** Use a nail to carefully poke a small hole in one corner of the bottom of each plastic bottle.
- » **Label the bottles and cups:** Use a grease pencil to label the plastic bottles with the name of the material you will put in it: Gravel, Sand, and Clay. Also, label your four collection cups to correspond with each bottle.
- » **Prepare the layers:** Place about two inch of gravel in the bottom of the Gravel bottle. Add two inches of coarse sand to the Sand bottle. Put two inches of clay in the Clay bottle.
- » **Set up the experiment:** Place each bottle upright over its corresponding collection cup, making sure the hole is positioned over the centre of the cup.
- » **Measure the water:** Measure 500 ml of water in your measuring cup.
- » **Start the test:** Pour the 500 ml of water into the bottle containing the gravel and start the stopwatch immediately.
- » **Record the time:** Stop the stopwatch when water first starts dripping out of the hole into the collection cup. Record the time on the data sheet below.
- » **Repeat the test:** Repeat steps 5-7 for the sand, soil, and clay bottles, using 500 ml of water for each test. Be sure to record the results for each material.

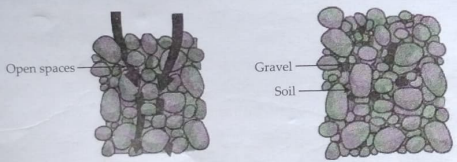


Bottle Filled with	Prediction	Observation
	Seepage of water (Very slow/Slow/Fast)	Seepage of water (Very slow/Slow/Fast)
Bottle-1 (Gravel)		
Bottle-2 (Sand)		
Bottle-3 (Clay)		

What are your findings? Do your findings match your predictions?

In this activity, you have observed that water seeps fastest through the gravel, slower through sand and slowest through clay. Why does this happen?

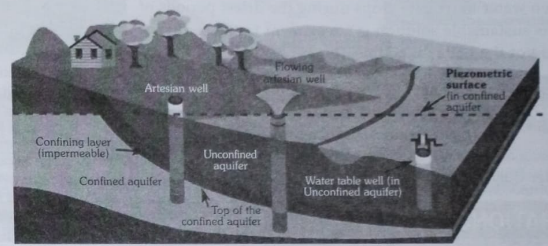
The space between the gravel's particles are wider than sand and clay. Hence, water can seep through the gravel more easily. Therefore, groundwater occurs almost everywhere beneath the land surface. Infiltration is the downward movement of water into the soil. Water infiltrate more readily if the spaces between the soil and rock particles are wider, open and interconnected with each other.



This is the continuous flow of this water through the spaces between soil and rock particles. As water enters, the soil temporarily stores it, making it available for plant roots and soil organisms.

Groundwater: The water that seeps down and fills underground spaces and rock fractures below the water table. Groundwater can be found almost everywhere. The water table may be deep or shallow; and may rise or fall depending on many factors. Heavy rains or melting snow may cause the water table to rise, or heavy pumping of groundwater supplies may cause the water table to fall.

Aquifers: These are typically made up of gravel, sand, sandstone, or fractured rock, like limestone. Water can move through these materials because they have large connected spaces that make them permeable. This water flows through pores and cracks and can collect as groundwater in underground layers called aquifers. The speed at which groundwater flows depends on the size of the spaces in the soil or rock and how well the spaces are connected.



Wells: We get water from underground by drilling wells or collecting it from springs. Groundwater is also often withdrawn for agricultural, municipal, and industrial use by constructing and operating extraction wells. Wells are holes drilled into an aquifers that are equipped with pumps to bring water to the surface. This water may be a few meters to hundreds meters below the ground, depending on the location. Springs are natural outlets where groundwater emerges at the surface, which can then be collected.

Groundwater Depletion: The groundwater of earth is not a unlimited source of water. It is a limited source of water that makes up about 30% of the world's freshwater, making it the second-largest reservoir of freshwater after ice caps and glaciers, which hold about 68.7%. While groundwater is less abundant than glaciers, it is significantly more plentiful than the freshwater found in lakes and rivers. Groundwater depletion is the long-term decline in the water table caused by excessive pumping of underground water from aquifers for agricultural use, industry, and domestic purposes, which exceeds the rate of natural replenishment. This leads to consequences like wells running dry, land subsidence, saltwater intrusion, and harm to aquatic ecosystems.

Groundwater depletion in "urban areas" is caused by a cycle of increased demand and reduced natural replenishment. Dense urban areas with extensive impervious surfaces like concrete and pavement prevent rainwater from soaking into the ground and recharging underground aquifers. Meanwhile, the high concentration of people, industries, and commercial activity leads to a dramatic increase in water extraction.



Water conservation practise in Ladakh region: During the spring season in Ladakh, streams often dry up, leading a water scarcity of water in the region. An ice stupa works by freezing

all in your own words!

water into a large conical shape during winter, storing it for gradual melting in the spring to provide water for crops. Water is piped from a stream, and at night, when temperatures drop, the water is sprayed through a pipe, freezing into a tower that grows taller over weeks. Ice stupas play a crucial role in supporting the local community by storing frozen water to irrigate crops during the driest periods of the year. This sustainable solution is a testament to the resilience and ingenuity of the Ladakhi people.



DO YOU KNOW?

Water is a renewable resource, but if we use it faster than nature can replace it, it may run out locally. Renewable resources like sunlight, wind and forests can also be replenished naturally if used wisely. Conserving water helps keep our groundwater truly renewable.

ILLUSTRATIONS

9 The water table varies from place to place. Why?

Ans.: Water table is the upper level of the Earth's layer where water is found. The water found below the water table is known as groundwater. The availability of groundwater or the level of water table depends on the amount of rainfall and recharge conditions in an area, e.g., in places near the river, one may obtain water after digging only

a few feet, while one may have to dig several feet down in a desert before you get water. Thus, the water table varies from place to place.

10 There is abundance of water on our planet Earth. Why is there a scarcity of water then?

Ans.: Although there is abundance of water on our planet Earth, but only a very small fraction of water is fit for consumption. More than 97% of water that exists on the Earth is in seas and oceans that is highly salty and unfit for drinking or other purposes.

Formulas

$$\begin{aligned} C \rightarrow K &= K = C + 273.15 \\ K \rightarrow C &= C = K - 273.15 \\ F \rightarrow C &= C = (F - 32) \times 5/9 \\ C \rightarrow F &= F = C \times (5/9) + 32 \\ F \rightarrow K &= K = (F - 32) \times 5/9 + 273.15 \\ K \rightarrow F &= F = (K - 273.15) \times 9/5 + 32 \end{aligned}$$

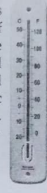
CONCEPT MAP

- Water cycle:** The continuous movement of water through processes like evaporation, condensation and precipitation.
- Seepage and groundwater:** Water infiltrates soil filling underground aquifers and moving through porous layers.
- Groundwater depletion:** Overrule of groundwater causing depletion leading to issues like dry wells and land subsidence.

Thermometer

It is an instrument used to measure temperature. It works on the principle of expansion of liquid on heating. Temperature is measured in Celsius and Fahrenheit scale for clinical and digital purpose but for scientific work, Kelvin scale is used. Different types of thermometer:

- Clinical thermometer
- Laboratory thermometer
- Digital thermometer



Temperature

Temperature is the degree of hotness and coldness of an object.

- Heat is a form of energy that produces feeling of hotness or coldness of an object. This effect is known as temperature.
- 1 cal = 4.2 J
- Any matter becomes warmer by gaining heat and cooler on releasing heat causing change in temperature.

Thermometer scales are Celsius ($^{\circ}\text{C}$), Fahrenheit ($^{\circ}\text{F}$) and Kelvin (K)

- Relation between three scales

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} = \frac{T_K - 273}{373 - 273}$$

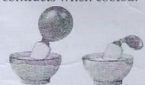
Heat Transfer in Nature

Effect of Heat on Matter

Types of Heat Transfer

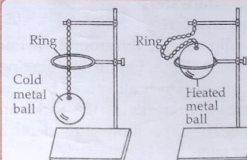
Matter Contracts when Cooled

Matter contracts when cooled.



Matter contracts on cooling, causing decrease in volume or shape and size of the matter.

Matter Expands on Heating



Matter (solid, liquid and gas) expands on heating, causing change in shape and size or volume of the matter.

Conduction

Transfer of heat without movement of solid material itself is known as conduction of heat.



Convection

Transfer of heat with the movement of molecules of fluid is known as convection of heat.



Radiation

Transfer of heat energy between objects without any medium is known as heat radiation.



Solved Examples

1. What differences can you find between a clinical thermometer and a laboratory thermometer?

Ans.: The clinical thermometer is smaller. It measures only from 35°C to 42°C. It is mainly used for medical purposes. There is a small bend or kink in the capillary tube. This prevents the mercury from coming down, when the thermometer is taken out of patient's mouth. Thus the doctor can read the temperature after removing the thermometer.

The laboratory thermometer is larger than clinical thermometer and it is used to measure temperature from 0°C up to 100°C or more in the laboratory. It has no kink in its capillary tube.

2. (i) Raghav has a temperature of 40°C. What would you suggest him to do?

(ii) Can you say which of these is the temperature on the surface of the Sun?

(a) 50°C (b) 500°C (c) 3000°C (d) 5500°C

Ans.: (i) Raghav has fever as his body temperature is more than the normal value (37°C) and he should see a doctor without any delay.

(ii) (d) : 5500°C

3. Dhruv got a slab of chocolate from his friend on his birthday. He kept it in his bag. On reaching home, he opened his bag and found a brown sticky liquid oozing out of the wrapper. It had spread over some of his books. What do you think happened?

Ans.: The chocolate melted due to the heat. It changed from a solid matter to a thick liquid that leaked out, and had spread over Dhruv's books.

4. Arrange these in decreasing order based on their expansion on heating: steel, milk, air.

Ans.: Air, milk, steel.

5. Can convection take place in solids? Why?

Ans.: No. The molecules in a solid are only free to vibrate about their fixed positions. For convection to take place, the molecules need

to move carrying the heat with them. Hence convection can not take place in solids.

6. In winters, when the Sun suddenly goes behind the clouds we feel cold. Can you say why?

Ans.: The clouds cut off the radiant heat from the Sun.

7. Why do fire fighters wear special shiny suits when they enter a burning building?

Ans.: The shiny suits reflect heat. This protects them from fire when they enter a burning building.

8. If you live on the top floor of a building, you must have noticed that, in summer the roof gets very hot. Suggest a few ways by which people who live on the top floor of a building can protect themselves from the radiant heat of the Sun?

Ans.: The roof top may be painted white so as to reflect the radiant heat from the sun.

If green or white Perspex sheets are used on a roof they prevent the main roof from absorbing the radiant heat of the Sun.

Watering the roof and maintaining a rooftop garden are the other ways to protect from the heat of the Sun.

9. (a) What is water table?

(b) What is groundwater?

(c) What is an aquifer?

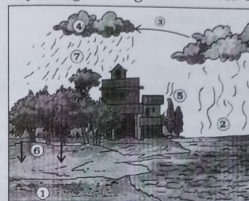
Ans.: (a) If we dig a hole in the ground, we find moist soil at certain depth. On further digging, we reach a depth at which all spaces between the soil particles and cracks in the rocks are filled with water. The upper limit of this depth is called the water table.

(b) The water found beneath the earth's surface and below the water table is called groundwater.

(c) At places beneath the earth's surface, the groundwater is stored between layers of hard

rocks below the water table. This is known as an aquifer. Water in the aquifers can be pumped out with the help of tubewells or hand pumps.

10. Study the given figure of water cycle.



(a) Name the number labels as shown in the figure.

(b) Define the process labelled as 5.

Ans.: (a) (1) Groundwater

(2) Evaporation (3) Condensation

(4) Clouds (5) Transpiration

(6) Infiltration (7) Precipitation

(b) The process labelled as 5 is transpiration. Transpiration is the loss of water in the form of water vapour from the aerial parts of a plant such as leaves.

11. Convert 25 °C to °F.

Ans.: $T(^{\circ}\text{F}) = \left(\frac{9}{5} \times T(^{\circ}\text{C})\right) + 32$

$T(^{\circ}\text{F}) = \left(\frac{9}{5} \times 25\right) + 32 = (9 \times 5) + 32 = 45 + 32 = 77^{\circ}\text{F}$

12. Convert 20 K into °C.

Ans.: $T(\text{K}) = 273 + T(^{\circ}\text{C})$

$20 = 273 + T(^{\circ}\text{C})$

$T(^{\circ}\text{C}) = 20 - 273 = -253^{\circ}\text{C}$

13. You may have noticed that a few sharp jerks are given to clinical thermometer before using it. Why is it done so? (NCERT Exemplar)

Ans.: A clinical thermometer has a kink which prevents immediate backflow of the mercury from the tube to the bulb, thus allowing us to read the temperature conveniently. This is the reason why we give a jerk to the thermometer

before taking a new reading. The jerk positions the mercury below the kink into the bulb.

14. At a camp site there are tents of two shades – one made with black fabric and the other with white fabric. Which one will you prefer for resting on a hot summer afternoon? Give reason for your choice. Would you like to prefer the same tent during winter?

(NCERT Exemplar)

Ans.: (i) On a hot summer afternoon, the tent made up of white fabric will be preferred as white colour is a bad absorber and good reflector of heat.

(ii) No, the black fabric tent will be preferred during winter.

15. Why is it advised not to hold the thermometer by its bulb while reading it?

(NCERT Exemplar)

Ans.: It is advised not to hold the thermometer by its bulb while reading it because on holding a thermometer by its bulb, the mercury in the bulb expands due to our body temperature and gives false reading.

16. For setting curd, a small amount of curd is added to warm milk. The microbes present in the curd help in setting if the temperature of the mixture remains approximately between 35°C to 40°C. At places, where room temperature remains much below the range, setting of curd becomes difficult. Suggest a way to set curd in such a situation. (NCERT Exemplar)

Ans.: (i) In order to maintain the desired temperature of the mixture, the container can be wrapped either by woollen material or any other poor conductor of heat. Alternatively, the mixture can be kept in a heat resistant container.

(ii) The container can be kept in the Sun or near the gas stove while cooking.

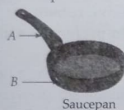
17. While constructing a house in a coastal area, in which direction should the windows preferably face and why? (NCERT Exemplar)

Ans.: The windows of houses in coastal areas should preferably face towards the sea as sea breeze will keep it cool during day time.

NCERT Section

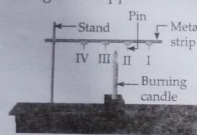
1. Choose the correct option in each case.

- (i) Your father bought a saucepan made of two different materials, A and B, as shown in figure. The materials A and B have the following properties—



Saucepan

- (a) Both A and B are good conductors of heat.
(b) Both A and B are poor conductors of heat.
(c) A is a good conductor and B is a poor conductor of heat.
(d) A is a poor conductor and B is a good conductor of heat.
(ii) Pins are stuck to a metal strip with wax and a burning candle is kept below the rod, as shown in figure. Which of the following will happen?



Heat transfer set-up

- (a) All the pins will fall almost at the same time.
(b) Pins I and II will fall earlier than pins III and IV.
(c) Pins I and II will fall later than pins III and IV.
(d) Pins II and III will fall almost at the same time.
(iii) A smoke detector is a device that detects smoke and sounds an alarm. Suppose you are fitting a smoke detector in your room. The most suitable place for this device will be:

- (a) Near the floor
(b) In the middle of a wall

- (c) On the ceiling
(d) Anywhere in the room

2. A shopkeeper serves you cold lassi in a tumbler. By chance, the tumbler had a small leak. You were given another tumbler by the shopkeeper to put the leaky tumbler in it. Will this arrangement help to keep the lassi cold for a longer time? Explain.

3. State with reason(s) whether the following statements are True [T] or False [F].

- (i) Heat transfer takes place in solids through convection. []
(ii) Heat transfer through convection takes place by the actual movement of particles. []
(iii) Areas with clay materials allow more seepage of water than those with sandy materials. []
(iv) The movement of cooler air from land to sea is called land breeze. []

4. Some ice cubes placed in a dish melt into water after sometime. Where do the ice cubes get heat for this transformation?

5. A burning incense stick is fixed, pointing downwards. In which direction would the smoke from the incense stick move? Show the movement of smoke with a diagram.

6. Two test tubes with water are heated by a candle flame as shown in figure. Which thermometers (Figure (a) or figure (b)) will record a higher temperature? Explain.

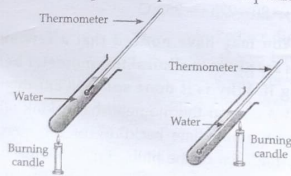


Fig. (a)

Fig. (b)

Two thermometers dipped in two test tubes

7. Why are hollow bricks used to construct the outer walls of houses in hot regions?
8. Explain how large water bodies prevent extreme temperature in areas around them.

9. Explain how water seeps through the surface of the Earth and gets stored as groundwater.
10. The water cycle helps in the redistribution and replenishment of water on the Earth. Justify the statement.

Exercise



Multiple Choice Questions

LEVEL - 1

Heat, Temperature and Conduction

1. A thermometer works on the principle of
(a) linear expansion of solid
(b) cubical expansion of solid
(c) uniform expansion of volume of liquid or gas with temperature
(d) Both (b) and (c).

[R]

2. Name the thermometer which ranges from -10°C to 140°C .

- (a) Clinical thermometer
(b) Digital thermometer
(c) Laboratory thermometer
(d) All of these

[R]

3. Which one of the following scale has lower fixed point at 0°C ?

- (a) Kelvin scale
(b) Fahrenheit scale
(c) Celsius scale
(d) All of these

[R]

4. Paheli and Boojho measured their body temperature. Paheli found her's to be 98.6°F and Boojho recorded 37°C . Which of the following statement is true?

- (a) Paheli has a higher body temperature than Boojho.
(b) Paheli has a lower body temperature than Boojho.
(c) Both have normal body temperature.
(d) Both are suffering from fever.

(NCERT Exemplar)

5. Why a digital thermometer is widely used for the measurement of the temperature?

- (a) Analog display
(b) Mercury
(c) High accuracy
(d) Low accuracy

[U]

6. When we heat one end of an iron rod, its other end also gets heated. Can you say, which one of the following is behind this?

- (a) Convection of heat
(b) Radiation of heat
(c) Insulation of heat
(d) Conduction of heat (NCERT Page 90)

[U]

7. When you heat the water in a pot, it boils. What do you infer from the observation?

- (a) Heat is a form of energy.
(b) Water can boil itself.
(c) Water develops heat on its own.
(d) None of these.

[U]

8. In which of the given option chemical energy is converted into heat energy?

- (a) Heater
(b) Refrigerator
(c) Candle
(d) Motor

[U]

9. Iron, hydrogen peroxide and carbon dioxide. When same amount of heat is supplied to these three substances, which expands most?

- (a) Hydrogen peroxide
(b) Carbon dioxide
(c) Iron
(d) All of these.

[An]

10. A liquid changes into a gas at a constant temperature known as its

- (a) absolute zero
(b) boiling point
(c) evaporation point
(d) dew point.

[R]

11. Petrol storage tanks are not painted black because

- (a) black is good conductor of heat
(b) black is bad conductor of heat
(c) black is good reflector of radiation
(d) black is good absorber of radiation.

(Page 91) **U**

12. In solid substances, heat is transferred by
(a) Conduction (b) Convection
(c) Radiation (d) (a) and (c)

(Page 91) **R**

13. The handles of cooking vessels are covered with plastic or wood because
(a) they are beautiful
(b) it is customary
(c) they are good conductors of heat
(d) they are bad conductors of heat.

(Page 91) **U**

14. In conduction, heat flows from
(a) hotter to hotter region
(b) colder to hotter region
(c) hotter to colder region
(d) colder to colder region.

(Page 91) **R**

15. Figure (a-d) shows the readings on four different thermometers. Indicate which of the reading shows the normal human body temperature?



(NCERT Exemplar)

16. The measure of degree of hotness or coldness of a body is called
(a) heat energy
(b) celsius
(c) kelvin
(d) temperature.

R

17. Which of the following is not a scale of temperature?
(a) Kelvin scale
(b) Celsius scale
(c) Fahrenheit scale
(d) Richter scale

R

18. The difference between lower fixed point and upper fixed point is divided into _____ parts on a Celsius scale.

- (a) 100 (b) 273
(c) 180 (d) 50

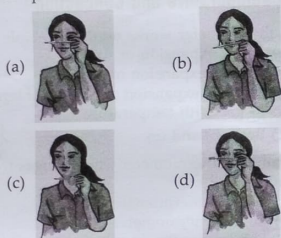
R

19. When two objects are in physical contact, the heat is transferred by

- (a) conduction
(b) convection
(c) radiation
(d) none of these.

(Page 91) **R**

20. Figure (a-d) shows a student reading a doctor's thermometer. Which of the figure indicates the correct method of reading temperature?



(NCERT Exemplar)

21. A polished, silvery surface is a
(a) good absorber and good reflector of heat
(b) good absorber and bad radiator of heat
(c) poor absorber and good reflector of heat
(d) poor reflector and good radiator of heat.

(Page 91) **R**

22. Substances which allow heat to pass through them are called
(a) conductors (b) insulators
(c) moderators (d) none of these.

(Page 91) **R**

23. A coin is dipped in the molten wax in a glass tube. If we heat the upper part of the glass tube, the wax around the coin will not melt because
(a) wax has a very high melting point
(b) wax is a good conductor of heat

- (c) glass is a good conductor of heat
(d) air and glass are bad conductors.

(Page 91) **U**

Convection

24. When you pour hot tea suddenly into a glass tumbler, it cracks. Because
(a) glass cannot resist heat.
(b) glass expands unevenly around the place of contact with the tea.
(c) tea has chemicals which react with glass tumblers.

- (d) the reason is yet to be discovered by the scientists.

(Page 94) **U**

25. Warm air is
(a) lighter than cold air
(b) heavier than cold air
(c) both have equal weights
(d) cannot be said.

(Page 94) **R**

26. Convection of heat takes place in

- (a) metals only
(b) liquids only
(c) gases only
(d) liquids and gases.

(Page 94) **R**

27. When you heat a bowl of water, you can observe that water starts circulating. This demonstrates
(a) conduction (b) convection
(c) radiation (d) none of these.

(Page 94) **U**

28. Birds glide effortlessly in the air with the help of
(a) conduction of heat in the air
(b) radiation of light through the atmosphere
(c) convection currents of air
(d) none of these.

(Page 91) **U**

29. The phenomenon involved in sea breeze and land breeze is
(a) convection (b) conduction
(c) radiation (d) none of these.

(Page 91) **R**

Radiation and Water Cycle

30. Process of change of state from gaseous state to liquid state is called

- (a) freezing
(b) sublimation
(c) boiling
(d) condensation.

(Page 98) **R**

31. When clouds get too heavy to hold water in the water cycle, what happens?
(a) The water falls to the Earth.
(b) The water evaporates.
(c) Another cloud forms.
(d) The clouds get very large.

(Page 98) **U**

32. Studying the water cycle, where is the purest water on Earth?
(a) Glaciers and ice caps
(b) Rivers and streams
(c) Sulfur springs
(d) Lakes and ponds

(Page 98) **An**

33. What turns water on the Earth into vapor in the water cycle?
(a) The Sun
(b) The Moon
(c) Global warming
(d) Water treatment centers

(Page 98) **U**

34. What is tapped into when digging a well looking for water in the water cycle?

- (a) An underground aquifer
(b) An underground river
(c) A sulfur spring
(d) An irrigation ditch

(Page 100) **R**

35. Select the option containing only renewable resources.

- (a) Fossil fuels, metals
(b) Groundwater, fossil fuels
(c) Forests, petroleum
(d) Groundwater, sunlight

(Page 100) **R**

36. Water can be obtained from aquifer by
(a) digging a tubewell
(b) hand pump
(c) infiltration
(d) either (a) or (b).

(Page 100) **R**

37. Which of the following is not a consequence of water scarcity?
(a) Decreased crop production
(b) Occurrence of water-borne diseases
(c) Drought
(d) Greenhouse effect

(Page 100) **U**

38. Which of the following is the way to conserve water?
- Mending the leaking taps.
 - Planting more and more trees.
 - Watering the plants in morning and evening hours.
 - All of these.

(Page 101) **Ap**

LEVEL - 2

39. Four bowls of water kept in different places
- In sunlight
 - Under a fan
 - In a closed cupboard
 - Outside at night
- After two days, bowl A has the least water left. What does this show?

- The sun helps water evaporate faster.
- Water changes into ice in sunlight.
- Air stops evaporation.
- The fan makes water condenses.

40. Aluminium is a better conductor of heat than wood. It is so because

- aluminium has a higher density than wood.
- aluminium has a higher specific heat capacity than wood.
- aluminium has more free electrons than wood.
- aluminium has a higher mass than wood.

An

41. Which of the following statements about radiation is/are true?

- Radiation can pass through solids.
 - Radiation can pass through liquids.
 - Radiation can pass through gases.
- 3 only
 - 2 and 3 only
 - 1 and 3 only
 - 1, 2 and 3

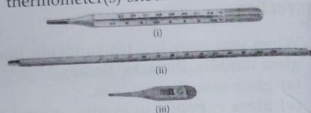
R

42. A dog pants with its tongue sticking out on a hot day. The reason is

- it is its habit
- producing more saliva cools down the body temperature
- saliva vapourizes, cooling the tongue
- it is a genetic disease found in few animals like dogs.

U

43. Boojho has three thermometers as shown in figure. He wants to measure the temperature of his body and that of boiling water. Which thermometer(s) should he choose?



- Thermometer (i) or (iii) for measuring body temperature and (ii) for measuring the temperature of boiling water.
- Thermometer (i) for measuring temperature of both.
- Thermometer (ii) for measuring temperature of both.
- Thermometer (iii) for measuring temperature of both.

(NCERT Exemplar)

44. Container A and container B are filled with equal amount of boiling water and the temperature of the water in the containers is measured with a thermometer some time later. It is observed that container A has a much lower temperature than container B. What are the possible reason(s)?

- Container A is black and container B is silver.
- Container A has a lid and container B is not covered.
- Container A is made of steel and container B is made of clay.

- 1 and 2 only
- 1 and 3 only
- 2 and 3 only
- 1, 2 and 3

An

45. Which of the given statements is true?

- Sea breeze happens during night time when the sea is cooler than the land.
- Sea breeze happens during day time when the land is cooler than the sea.
- Land breeze happens during night time when the land is cooler than the sea.
- Land breeze happens during day time when the sea is cooler than the land.

U

46. When a solid melts, the temperature remains constant even though heat is being supplied because

- the heat is being used to break up the intermolecular bonds.
- the solid is not absorbing any heat.
- the molecules are moving faster.
- the molecules are far apart.

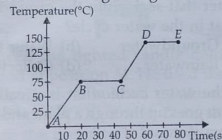
U

47. On a cold day, it is hard to open the lid of a tight container. But when you gently heat the neck you can easily open the lid. Why?

- On heating glass expands and lid contracts.
- On heating lid expands more than the neck and thus slides easily.
- Neck becomes slippery on heating.
- Lid of the bottle cannot bear the heat.

U

48. Which point on the given graph shows that the substance is beginning to boil?



- B
- C
- D
- E

An

49. When some liquid evaporates, the average speed of the remaining molecules will
- increase because the more energetic molecules have left
 - decrease because the more energetic molecules have left
 - remain unchanged because all molecules have about the same speed
 - increase because there are fewer molecules.

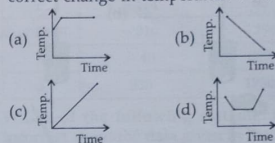
R

50. Cooling always accompanies evaporation because

- the air molecules cool the liquid surface.
- the more energetic molecules leave the liquid.
- there are fewer liquid molecules left in the liquid.
- the escaped molecules return to the liquid.

R

51. A pot full of water is put to boil on the stove. Which one of the graphs shows the correct change in temperature over time?



An

52. Amit found that a glass P was stuck inside another glass Q. What could he do to separate the two glasses?

- Pour hot water into glass P and put glass Q in hot water.
- Pour hot water into glass P and put glass Q in cold water.
- Pour cold water into glass P and put glass Q in cold water.
- Pour cold water into glass P and put glass Q in hot water.

R

53. Two blocks of ice when pressed together join to form one block. This happens because

- Melting point rises with pressure
- Melting point falls with pressure
- Heat is absorbed from outside
- Heat is rejected to outside.

U

54. There is a hole in the middle of a copper plate. When heating the plate, diameter of hole would

- always increase
- always decrease
- remains the same
- none of these

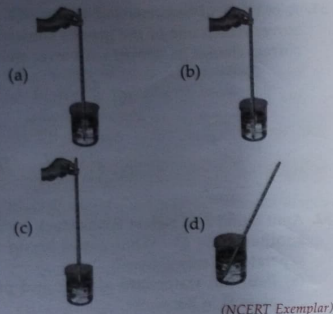
U

55. In which of the following process convection does not take place primarily?

- Sea and land breeze
- Trade wind
- Boiling of water
- Warming of glass of bulb due to filament

U

56. Four arrangements to measure temperature of ice in beaker with laboratory thermometer are shown in figure (a, b, c and d). Which one of them shows the correct arrangement for accurate measurement of temperature?



(NCERT Exemplar)

57. On heating a substance which of the following physical quantity changes?

- (a) Density (b) Mass
(c) Volume (d) Both (a) and (c)

R

58. Cooling of a motor car is done by

- (a) Conduction (b) Convection
(c) radiation (d) All of the given

R

59. Water is an inexhaustible, renewable resource, still there is shortage of water at many places of the world. It is because

- (a) much water available to us is not in the usable form
(b) 97.4% of all available water is frozen in glaciers and polar ice caps
(c) we cannot make use of water that is present in rivers and lakes
(d) we cannot make use of rainwater.

60. X of the total available water is present in seas and oceans, about Y water is frozen in glaciers and in the polar icecaps, and the remaining Z is available as freshwater in the liquid form.

- | X | Y | Z |
|-----------|------|------|
| (a) 60% | 38% | 2% |
| (b) 97.4% | 0.6% | 2% |
| (c) 97.4% | 2% | 0.6% |
| (d) 70.5% | 20% | 9.5% |

61. Which stage is NOT part of the water cycle?
(a) Emancipation (b) Collection
(c) Evaporation (d) Precipitation

62. Studying the water cycle, why is the ocean salty?

- (a) Groundwater picks up salt and minerals as it makes its way to the ocean.
(b) It is rinsed from the scales and skin of ocean animals.
(c) Salt is carried down in rainwater.
(d) Human pollution made it that way.

63. Considering the water cycle how old could the water be in glaciers and ice caps?

- (a) Millions of years
(b) Billions of years
(c) Thousands of years
(d) Hundreds of years

64. Water that soaks into the ground is called what in the water cycle?

- (a) Groundwater (b) Water table
(c) Rainwater (d) Collected

65. In the water cycle what is it called when water goes up through a plant and is turned into a vapour?

- (a) Evaporation (b) Precipitation
(c) Transpiration (d) Condensation

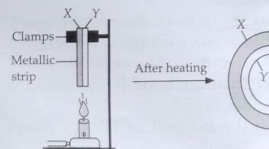
66. Condensation is when warm vapor rises, gets cold and does what in the water cycle?

- (a) Gathers in clouds.
(b) Falls out of the sky.
(c) Warms up again.
(d) Keeps rising until it turns to ice.

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

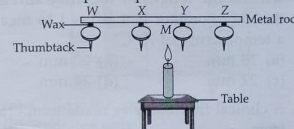
67. Aparna heated a bi-metallic strip made of two different types of metals for some time.



After heating it, she observed that the metallic strip curved as shown in the figure. Which of the following would best explain the conclusion derived by Aparna about the experiment she had done?

- (a) Metal X is a better conductor of heat than metal Y.
(b) Metal X is a poorer conductor of heat than metal Y.
(c) Metal X is of better quality than metal Y.
(d) Metal X is more flexible than metal Y.

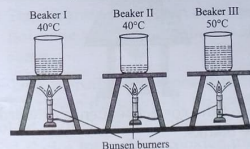
68. Rohan set up an experiment as shown here.



He placed the lighted candle below the metal rod at position M. He used the same amount of wax to hold all the thumbtacks at positions W, X, Y and Z on the rod. Arrange the thumbtacks according to the time they take to drop from the rod, from the first to the last.

- (a) Y, X, W, Z (b) Y, X, Z, W
(c) X, Y, W, Z (d) W, Z, Y, X

69. Three glass beakers of similar sizes are filled with different amounts of water heated to different temperatures.



Arrange the beakers in order, starting with the one with the maximum heat.

- (a) I, III, II (b) II, I, III
(c) III, I, II (d) III, II, I

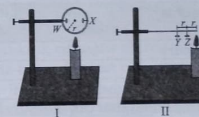
70. The melting and boiling points of four different substances are shown in the table. The room temperature is 30°C.

Substance	Melting point (°C)	Boiling point (°C)
P	-114	80
Q	-210	-196
R	-40	360
S	330	1750

Which of the following statements are incorrect about the data given in the table?

- (i) Substance P is in solid state at room temperature.
(ii) Substance Q is in liquid state at room temperature.
(iii) Substance R is gas at -150°C and solid at -220°C.
(iv) Substance S is in solid state at room temperature.
(a) (i) and (ii) only
(b) (i), (ii) and (iii) only
(c) (iii) and (iv) only
(d) (i), (ii), (iii) and (iv)

71. In the arrangements I and II shown in figure, four pins W, X, Y and Z are fixed to a metal loop and metal rod with the help of wax. Loop and rod are made of same material and equal thickness. Which set of pins will fall simultaneously? (r is the radius of the loop.)



- (a) Y and Z only (b) W, X and Y only
(c) W and X only (d) W, X, Y and Z

72. Ms. Rakhi, a science teacher listed some keywords on the blackboard as :

- (i) Freezing (ii) Evaporation (iii) Melting
(iv) Sublimation (v) Condensation (vi) Vaporisation

She asked students to match the keywords with the common changes which are listed as :

P : When we put a lid on a vessel containing hot milk, the inside of the lid gets wet.

- Q : When we breathe onto a cold mirror it turns foggy.
 R : When we spread hard butter on a warm toast, it becomes soft.
 S : Solid piece of an air freshener kept on the bathroom shelf disappears in a few days.
 T : When we put ice cubes in water, they disappear after some time.
 U : Milk left in the freezer becomes ice cream.

The correct match is

- (a) P - (ii); Q - (vi); R, S - (iii); T, U - (i)
 (b) P, Q - (ii); R - (iii); S - (iv); T, U - (i)
 (c) P, Q - (v); R, T - (iii); S - (iv); U - (i)
 (d) P - (i); Q, R - (v); S, T - (iii); U - (iv)
73. The diagram shows a man standing under a shelter on a sunny day. Given that the man feels hot, which of the following shows the processes of how thermal energy from the Sun reaches the man?



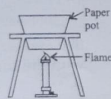
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74. Three liquids with their respective boiling and freezing points are shown in the given table.

Liquids	Freezing point	Boiling point
Mercury	-39°C	357°C
Alcohol	-112°C	78°C
Pentane	-180°C	36°C

A thermometer that can measure both -50°C and 38°C should be filled with _____.

- (a) Mercury only
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 (d) Either alcohol or pentane
75. Some steamboat restaurants use paper pots for their customers to heat the food themselves. What is the reason for the paper not to catch fire when in contact with the flame?



- (i) The paper is thin and therefore heat is conducted quickly to the water in the paper pot.
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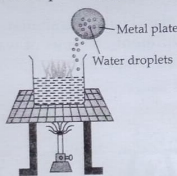
76. The length of mercury thread in a mercury thermometer is 2 mm at the ice point and 26 mm at the boiling point of water. What will be the length of mercury thread in this thermometer if it is used to measure a temperature of 75°C?

- (a) 18 mm (b) 20 mm
 (c) 22 mm (d) 24 mm

77. A clinical thermometer reads temperature from

- (a) 35°C to 42°C (b) -10°C to 110°C
 (c) 35°C to 45°C (d) 0°C to 100°C

78. The water in the beaker is _____ and is converted into _____. The process is known as _____. If you hold a metal plate near the steam, the steam _____ into water and water droplets can be seen on the plate. This is an example of _____ change.

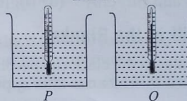


Select the correct sequence of words to fill the blanks.

- (a) Boiling, Steam, Vaporisation, Condenses Irreversible
 (b) Boiling, Vapours, Boiling, Converts Irreversible
 (c) Heating, Gas, Condensation, Changes Reversible
 (d) Boiling, Steam, Vaporisation, Condenses Reversible

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 (a) Conduction (b) Convection
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80. In the following experimental setup as shown in figure, equal volumes of boiling water were poured into the containers and the temperatures of both containers and contents were the same. After an hour the temperature of container P was found to be lower than that of container Q. This shows that P is a better _____.



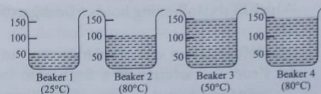
- (a) Conductor and radiator
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81. 1 litre water is poured into a small vessel and 10 litre water is poured into a big vessel. Heat is supplied to both vessels so that the water starts boiling. Then

- (a) Temperature of water in both vessels is same and the heat supplied is also same.
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82. Cooking utensils are supposed to transfer the heat from the source to the food being cooked. Which of the following is not a good choice for making cooking utensils?
 (a) Steel (b) Copper
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83. Karan performed an experiment in which he took four beakers containing different amount of water at different temperatures. He added equal amount of salt into each beaker.



Identify the beaker in which salt takes longer time to dissolve.

- (a) Beaker 1 (b) Beaker 2
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84. An iron disc at 50°C is put into a vessel full of water at 50°C. The heat will
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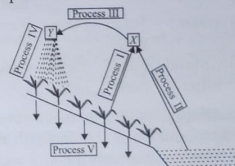
85. At room temperature the internal and external radii of a metal ring are r_1 and r_2 as shown in the figure. When the ring is heated at 100°C, then
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86. Which of the following statements is/are true?

- I. Dark coloured cloths are preferred during night.
 II. Sea breeze blows during day time.
 III. There is a kink in a digital thermometer and kink contains mercury.
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87. Refer to the given diagram showing a natural cyclic process.



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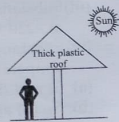
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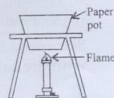
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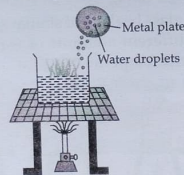
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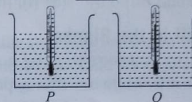
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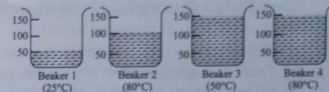
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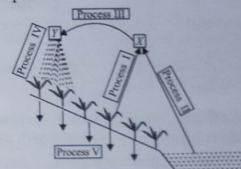


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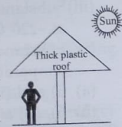


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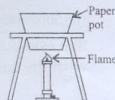
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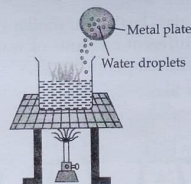
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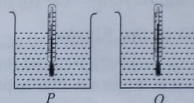


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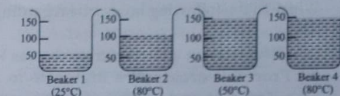
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 (c) Temperature of water in both vessels is not same but heat supplied is same.
 (d) Temperature of water in both vessels is not same and heat supplied is also not same.

82. Cooking utensils are supposed to transfer the heat from the source to the food being cooked. Which of the following is not a good choice for making cooking utensils?

- (a) Steel (b) Copper
 (c) Brass (d) Plastic

83. Karan performed an experiment in which he took four beakers containing different amount of water at different temperatures. He added equal amount of salt into each beaker.

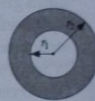


Identify the beaker in which salt takes longer time to dissolve.

- (a) Beaker 1 (b) Beaker 2
 (c) Beaker 3 (d) Beaker 4

84. An iron disc at 50°C is put into a vessel full of water at 50°C. The heat will
 (a) flow from iron disc to water
 (b) flow from water to iron disc
 (c) not flow from iron disc to water or vice versa
 (d) None of these

85. At room temperature the internal and external radii of a metal ring are r_1 and r_2 as shown in the figure. When the ring is heated at 100°C, then
 (a) r_1 decreases and r_2 decreases
 (b) r_1 decreases and r_2 increases
 (c) r_1 increases and r_2 decreases
 (d) r_1 increases and r_2 increases.

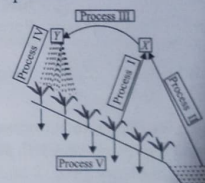


86. Which of the following statements is/are true?

- I. Dark coloured cloths are preferred during night.
 II. Sea breeze blows during day time.
 III. There is a kink in a digital thermometer and kink contains mercury.
 IV. The hotness of an object is determined by its temperature.

- (a) I and II only (b) II and III only
 (c) I, II and IV only (d) I, II, III and IV

87. Refer to the given diagram showing a natural cyclic process.



Which of the following holds true regarding it?

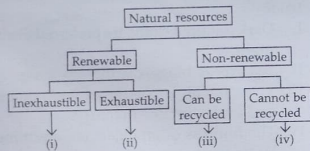
- Deforestation will decelerate process V.
- Process I occurs at a high rate in a rainforest and is absent in deserts whereas process III is involved in dew formation.
- The physical state of water at X is liquid and that at Y is gaseous.
- Both processes II and IV involve conversion of one physical state of water into another.

88. A group of social workers has transformed a dry area in the Rajasthan's Alwar district into a green place. They have revived dried-up rivers by constructing water-harvesting structures.

What are the names of those rivers?

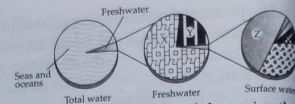
- Arveri and Ruparel
- Sarsa and Bhagani
- Arveri and Jahazwari
- All of these

89. Refer to the given flow chart and identify (i), (ii), (iii) and (iv).



- | | (i) | (ii) | (iii) | (iv) |
|-----|--------------|-------------|---------|-------------|
| (a) | Ground-water | Forest | Metal | Fossil fuel |
| (b) | Water | Forest | Metal | Fossil fuel |
| (c) | Water | Fossil fuel | Plastic | Forest |
| (d) | Ground-water | Fossil fuel | Metal | Forest |

90. Refer to the given diagrammatic representation of distribution of water on Earth and select the correct statements for X, Y and Z.



- X is a persistent body of dense ice that is constantly moving under its own weight.
 - Y is found below the water table and rainwater is its main source.
 - Z is an important source of water to the perennial rivers.
- (i) and (ii) only
 - (ii) and (iii) only
 - (i) and (iii) only
 - (i), (ii) and (iii)

Fill in the Blanks

- The water in the black can becomes _____ than that in white can after exposing to the Sun.
- In solids, generally, heat is transferred by _____.
- The boiling point of water on the Fahrenheit scale is _____.
- In liquids and gases, heat is transferred by the process of _____.
- In vacuum, heat energy can travel by the process of _____.
- Black colour is a _____ absorber of heat.
- The handles of cooking utensils are made of _____.
- Liquids are _____ of heat and gases are _____ of heat.
- A room heater heats all the air in a room by setting up _____ in the air.
- Digital thermometers do not use _____.
- Most of the water that evaporates on the Earth comes from _____.
- When water is heated, it changes from liquid into a gas called _____.
- As water goes around and around the water cycle, the amount of water on earth _____.
- Water that comes from plants and is then released into the air as water vapour through the process of _____.
- As water moves through the water cycle it changes _____.

True or False

- The temperature of every person may not be 37°C .
- The range of a laboratory thermometer is from 0°C to 110°C .
- Mercury is a toxic substance.
- Air is good conductor and water is poor conductor of heat.
- The transfer of heat by radiation does not require any medium.
- In summer, we prefer light-coloured clothes and in winter we usually wear dark-coloured clothes.
- During the day, land loses heat faster than water and becomes cooler.
- On heating, the density of the substance always decreases.
- The constant movement of water while heating is called convection current.
- Cooking of vegetables is an electrical change.
- Water boils at 212°F temperature but, it begins to evaporate at 32°F .
- Groundwater cannot be extracted using a water well.
- Mist is the tiny droplets of water hanging in the air.
- The sub-surface runoff is usually referred as interflow.

Match the Following

In this section, each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

- | List-I | List-II |
|-------------------|------------------------------------|
| (P) Heat | (1) Measure of hotness or coldness |
| (Q) Temperature | (2) Good absorber |
| (R) Black surface | (3) Form of energy |
- P-(3), Q-(1), R-(2)
 - P-(1), Q-(2), R-(3)
 - P-(3), Q-(2), R-(1)
 - P-(2), Q-(3), R-(1)

- | List-I | List-II |
|----------------------------------|---------------------------|
| (P) Actual movement of molecules | (1) Convection currents |
| (Q) Rubber, cork, plastic, wood | (2) 100°C |
| (R) Boiling point of water | (3) Good conductors |
| (S) Gold, silver, copper, iron | (4) Insulators |
- P-(1), Q-(4), R-(2), S-(3)
 - P-(4), Q-(2), R-(1), S-(3)
 - P-(1), Q-(2), R-(3), S-(4)
 - P-(3), Q-(1), R-(4), S-(2)

Assertion & Reason Type

Directions : In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as:

- If both assertion and reason are true and reason is the correct explanation of assertion
- If both assertion and reason are true but reason is not the correct explanation of assertion
- If assertion is true but reason is false
- If assertion is false but reason is true.

- Assertion :** Temperature is the measure of the heat energy.

Reason : Energy is the capacity to do work.
- Assertion :** Convection is a process of transfer of heat in which actual movement of material medium required.

Reason : The heat from the sun reaches us through millions of miles of empty space by convection.

3. **Assertion** : Heat energy is transferred from one body to another due to a temperature difference between them.

Reason : Heating a substance causes a rise in temperature.

4. **Assertion** : When a very hot liquid is poured into a thick glass tumbler it cracks.

Reason : Unequal expansion of the inner and outer glass walls causes the glass to crack.

5. **Assertion** : Conservation may be defined as the controlled utilization of natural resources for the benefit of all life so that it may yield sustainable benefits to the present as well as future generations.

Reason : Industrial processes add chemicals and metals to the groundwater, thereby increasing the quality of groundwater.

6. **Assertion** : Water table at a particular place remains constant always.

Reason : The upper limit of the Earth's layer where water is found is called an aquifer.

7. **Assertion** : Due to increased urbanization, less amount of water is seeping into the ground, thereby causing lowering of water table.

Reason : Increased urbanization has led to deforestation and construction of houses, shops, offices, roads and pavements.

8. **Assertion** : Deep groundwater is usually safe for drinking.

Reason : As groundwater passes through the soil, it gets filtered by several layers of sand and rocks, and becomes almost free of suspended impurities.

9. **Assertion** : Rainwater harvesting increases water availability as well as checks the declining of water table.

Reason : Rainwater harvesting improves the quality of groundwater and is environment friendly.

10. **Assertion** : Surface water is the rainwater that has not yet seeped into the ground or gone into the sea.

Reason : The continuous cycling of water among its three forms keeps the total amount of water on the earth constant even when the whole world is using it.

Comprehension Type

PASSAGE-I : An iron rod is heated by a burning candle. Iron rod is a good conductor of heat because heat is transferred from one end to other end. The property of the conductor is called conduction.

Copper is the conductor of heat because it transfers heat from one end to other.

Steel is the conductor of heat, therefore, utensils made of steel are widely used for the preparation of food.

Aluminium is also a good conductor of heat and therefore, used as utensils for the food preparation.

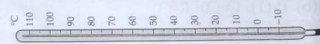
1. Which one of the following is not a good conductor of heat?

(a) Soil (b) Aluminium
(c) Tungsten (d) Steel

2. Why aluminium metal is used for making utensils for food preparation?

(a) Because it is a good insulator of heat.
(b) Because it is a good conductor of heat.
(c) Because it is good conductor of electricity.
(d) None of these.

PASSAGE-II : Laboratory thermometer is used for measuring the temperature in the science laboratory was shown in the figure. This thermometer cannot be used to measure the human body temperature.



Glass bulb containing mercury

1. Thermometers are used to measure the temperature. Thermometer senses the temperature by which one of the following liquids?

(a) Sulfur based fluid
(b) Potassium based fluid
(c) Hydrocarbon based fluid
(d) None of these.

2. What is the range of laboratory thermometers which are in the market?

(a) - 1 to 100 degree celsius
(b) - 10 to 1000 degree celsius
(c) -10 to 110 degree celsius
(d) None of these.

PASSAGE-III : We are aware that about 71% of the Earth's surface is covered with water. Almost all the water on the Earth is contained in the seas and oceans, rivers, lakes, ice caps, as groundwater and in the atmosphere. However, most of this water is not fit for human consumption directly. The water that is fit for use is freshwater. Around 2.5% of all water on Earth is freshwater. Only about 1% of the world's freshwater is readily available for human use, with most of the Earth's freshwater being inaccessible due to being trapped in glaciers, ice caps, and deep groundwater. Of the total amount of freshwater on the planet, roughly 30% is groundwater, while a significant portion is locked in ice. The vast majority of Earth's water (over 96%) is saltwater in oceans, which is not suitable for drinking without expensive desalinization.

1. How much is fresh water available on the Earth?

(a) 2.5% (b) 3.3%
(c) 4.3% (d) 5.5%

2. Most of the freshwater contained in which of the following sources of water?

(a) River (b) Lakes
(c) Groundwater
(d) Ice caps and glaciers.

3. How much fresh water is available on the Earth in the underground ?

(a) About 10% (b) About 30%
(c) About 20% (d) About 50%

Subjective Problems

Very Short Answer Type

- Give the name of instrument, which is used for measurement of heat.
- At which temperature does a liquid start to convert into gas?
- Name the method of heating of the earth by the sun.
- Convert 77°F to $^{\circ}\text{C}$.
- Give one condition for heat to flow between two bodies.
- Write one advantages of use of mercury in thermometer.
- What are insulators?
- Write the relation between Celsius and Kelvin scales.
- Name the thermometer which is used to measure the temperature of human body.
- What is boiling point of water?
- What are the various modes of transfer of heat?
- Which thermometer is used to measure the temperature of a furnace?
- What is the water cycle?
- What are the main stages of the water cycle?
- How does evaporation occur in the water cycle?
- What is condensation in the water cycle?
- How does precipitation happen in the water cycle?
- What factors influence the type of precipitation (rain, snow, sleet, hail) in the water cycle?
- Explain the process of runoff in the water cycle.
- What is infiltration in the water cycle?

- How does transpiration contribute to the water cycle?
- How does the Sun's energy drive the water cycle?
- Explain the role of clouds in the water cycle.
- What is the significance of groundwater in the water cycle?
- How does the water cycle regulate the Earth's temperature?
- Why is the infiltration of water low in the region of non-porous soil?
- Snowfall is common in the polar region and mountainous regions. Explain.
- Distinguish between evaporation and transpiration.
- How is groundwater recharged? Why is it important?
- Why is the handle of a kettle covered with plastic?

Short Answer Type

- Define the lower fixed point and upper fixed point on temperature scales. Give these points on different temperature scales.
- Write the differences between conduction, convection and radiation.
- What precautions should we take while using a clinical thermometer?
- Explain an activity to show how to measure temperature of water with laboratory thermometer.
- Why boiling water is not used to sterilize a clinical thermometer?
- What do you mean by conduction mode of transfer of heat energy?
- If difference of temperature between A and B is 5°C , what is the difference in Kelvin's scale?
- The temperature of a copper bar is 50°C . Find its temperature in Fahrenheit ($^{\circ}\text{F}$) and Kelvin (K).
- What are the factors affecting the heat lost or gained by a body?
- Sonia took four identical iron balls of temperature 6°C , 10°C , 20°C and 32°C . She kept them in physical (thermal) contact with one another. After some time, she measures the equilibrium temperature attained by those four balls. What is the value of equilibrium temperature?

Long Answer Type

- Explain with the help of an activity that silver is the best conductor of heat.
- Draw labelled diagram of a clinical thermometer and explain its construction.
- Explain conduction of heat in solids.
- What are the differences between heat and temperature?
- Neha is a student of science. She has participated in science exhibition organised by her school. In this science fair she demonstrated that of water and steam can exist in a same closed container at 100°C and water along with ice can exist in same container at 0°C . Explain how is it possible to have two different states of matter at same temperature.
- Explain the water cycle with a diagram.
- What is the aim of conservation of water? How can we conserve water?



Integer/Numerical Value Type

- The reading of Centigrade thermometer, coincides with that of Fahrenheit thermometer in a liquid. The temperature of the liquid is $-(5 \times n)^{\circ}\text{C}$. What is the value of n ?
- On a hilly region, water boils at $(15 \times n)^{\circ}\text{C}$. The temperature expressed in Fahrenheit is 194°F . What is the value of n ?

- A faulty thermometer has its fixed points marked as 5°C and 95°C . Temperature of a body as measured by the faulty thermometer is 59°C . The correct temperature of the body on Celsius scale is $\frac{120}{n}^{\circ}\text{C}$. What is the value of n ?



Case Based Questions

Case I :

Table : The upper and the lower fixed points on different temperature scales.

	Celsius Scale	Fahrenheit Scale	Kelvin Scale
Lower fixed point	0°C	32°F	$\sim 273 \text{ K}$
Upper fixed point	100°C	212°F	$\sim 373 \text{ K}$

To convert the temperature from one scale to another, the following relation is used:

$$\frac{\text{Temperature on one scale} - \text{Lower fixed point}}{\text{Upper fixed point} - \text{Lower fixed point}} = \frac{\text{Temperature on other scale} - \text{Lower fixed point}}{\text{Upper fixed point} - \text{Lower fixed point}}$$

$\therefore$ The relation between Celsius scale, Fahrenheit scale and Kelvin scale are given as :

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} = \frac{T_K - 273}{373 - 273}$$

CFQs

- Convert 20°C to $^{\circ}\text{F}$.
(a) 0°F (b) 67°F
(c) 86°F (d) 68°F
- Choose the incorrect formula.
(a) $T_F = \left(\frac{9}{5} \times T_C\right) + 32$ (b) $T_C = \frac{5}{9}(T_F - 32)$
(c) $T_K = T_C + 273$ (d) $T_C = \frac{9}{5}(T_F - 32)$
- Which of the following statements is/are true?
(a) Celsius ($^{\circ}\text{C}$) is the unit of temperature.
(b) $30^{\circ}\text{C} = 303 \text{ K}$.
(c) When heat energy flows from one body to another, the change in temperature of the bodies need not be equal.
(d) All of these.

- 0°C is the same temperature as
(a) 0°F (b) 222°F
(c) 32°F (d) 100°F
- At what temperature are the readings on the Celsius and the Fahrenheit scales the same?
(a) -40°C (b) -40°F
(c) 40°F (d) Both (a) and (b)

Case II : Heat flows from hotter objects to colder objects. When an object is at a temperature different from its surroundings, heat transfer takes place such that the body and the surroundings reach the same temperature. For example, when milk is boiled and the flame is off, the milk slowly transfers heat to the surroundings and becomes cooler. There are three modes of heat transfer: Conduction, convection and radiation.

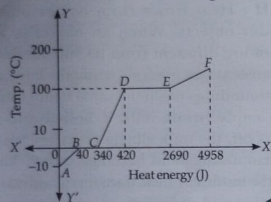


Figure: Heating by convection

- Which of the following is true in case of mode of transmission of heat?
(a) Convection is possible only in case of liquids and gases.
(b) Radiation is the fastest mode of heat transfer.
(c) Conduction is possible only in case of solids.
(d) All of the above.
- Heat flow from one body to another body stops when both bodies attain equal
(a) temperature (b) heat energy
(c) mass (d) volume.
- In a thermos flask, the loss of heat energy due to which of the following method is minimized?
(a) Conduction (b) Convection
(c) Radiation (d) All of these.

Case III : Heat energy can bring a change in the physical dimensions (length, area, volume), state and the chemical properties of a substance. Total energy of the particles increases as the temperature rises. The heating curve shows that the temperature stays constant until the solid is completely melted. The melting process requires a lot of energy.

The heating curve of a particular substance in solid state is as shown in the figure.



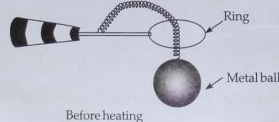
- The change of state in the graph is represented by
 - part AB
 - part BC
 - part DE
 - both (b) and (c).
- The amount of heat energy absorbed by the substance to change completely from liquid at its boiling point to gaseous state is
 - 1320 J
 - 2270 J
 - 2150 J
 - 1914 J
- If mass of the substance is 20 g, then the heat energy required to melt 1 g of the substance is
 - 30 J
 - 15 J
 - 60 J
 - 120 J

Case IV : All solids are found to expand on heating and contracts on cooling. This is called thermal expansion.

Thermal expansion is the tendency of matter to change in shape, area and volume in response to a change in temperature. When a solid is heated, its atoms vibrate faster about their fixed points. Metallic railway tracks have small gaps so that when the sun heats them, the tracks expand into these gaps and don't buckle.



- A thermometer works on the principle of
 - linear expansion of solid
 - cubical expansion of solid
 - uniform expansion of volume of liquid or gas with temperature
 - both (b) and (c).
- Bimetallic strip works on the principle of
 - unequal expansion of solids
 - unequal contraction of solids
 - equal expansion of solids
 - both (a) and (b).
- A metallic rod is bent in the form of a rectangle as shown in the given figure and heated. Then the gap between the ends A and B
 - increases
 - decreases
 - remains same
 - cannot be determined.
- From the given figure, conclude the correct option from the following.



- The solids contract on heating.
- Before heating, when the ball is kept on the ring it just slips through the ring.
- Solids expand on cooling.
- All of the above.

Case V : Clinical thermometer, which is used to measure our body temperature. It consists of a long, narrow, uniform glass tube. It has a bulb filled with mercury at one end. A constriction is provided, which does not allow mercury to fall back into the bulb.



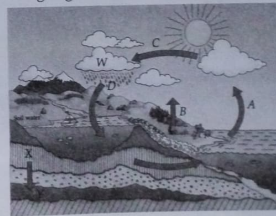
- Clinical thermometer is calibrated in
 - celsius scale
 - fahrenheit scale
 - absolute scale
 - both (a) and (b).
- Temperature is measured in
 - degree celsius
 - kelvin
 - degree fahrenheit
 - all of the above.
- S_1 : Clinical thermometer cannot be used to measure the temperature of melting ice.
 - S_2 : The range of clinical thermometer is from 35°C to 42°C.
 - Both S_1 and S_2 are correct.
 - S_1 correct and S_2 is incorrect.
 - Both S_1 and S_2 are incorrect.
 - None of them is correct.

Case VI : Four billion people face severe water scarcity for at least one month each year, and over two billion live in regions with inadequate water supply. By 2025, it was projected that 1.8 billion people would face "absolute water scarcity". Over two billion people live in countries where water supply is inadequate. Half of the world's population could be living in areas with water scarcity by 2025. These numbers highlight that water scarcity is a widespread global crisis affecting a significant portion of the world's population, a situation expected to worsen due to climate change and population growth. Climate change is one of the key drivers of the loss and degradation of freshwater ecosystems

and the unprecedented decline and extinction of many freshwater-dependent populations, particularly due to land use and pollution.

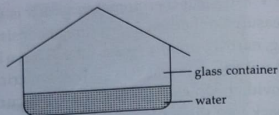
- According to Falken Mark, water stress occurs when available
 - water availability is less than 1000 cubic metre per person per year.
 - there is no water scarcity.
 - there is flood.
 - water availability is more than 1000 cubic metre per person per day.
- What are the causes for the scarcity of water in our urban areas?
 - Population Growth
 - Overuse and wastage
 - Unplanned urbanization
 - All of these
- What is the main cause of water pollution?
 - Acid rain
 - Hazardous Industrial waste
 - Climate change
 - Burning of fossil fuel

Case VII : Given cycle represents water cycle which involves constant exchange of water between the air, land and sea, and between the living organisms and their environments.



- Which of the following processes shows the conversion of water into gaseous state in the water cycle?
 - A and D
 - A and W
 - C and X
 - A and B.

2. Which among the following is responsible for liquification of evaporated water?
(a) D (b) A (c) X (d) B
3. Y is recharged through the process X. X is
(a) condensation
(b) evaporation
(c) infiltration
(d) precipitation.
4. Sally used a sealed glass container containing 300 ml of water as a model of the water cycle. She left the model in a garden on a cool night. When she looked at it the next morning, Sally found water droplets on the underside of the cover.



What do the water droplets found on the underside of the cover represent in the given water cycle?

- (a) C (b) Y (c) D (d) B
5. Which of the following represents the locked water?
(a) A (b) Y (c) D (d) B

Measurement of Time and Motion

CHAPTER

03

Learning Outcomes...

- Time
- Simple Pendulum
- Motion
- Distance and Speed
- Distance – Time graph
- Velocity – Time graph

Introduction

= collect be questions

= imp points

Time is the ongoing sequence of events in the past, present, and future that allows us to measure duration and the interval between occurrences. It is a concept that has been essential to human understanding since the beginning of ancient civilization. It is a measure of the duration of events and is commonly defined by the natural phenomena that happens over and over again in a regular fashion. Therefore, day and night were probably the first natural phenomenon to which humans were exposed in a cyclical manner. Thus, day and night were first used to define time. The time interval between the first noon and the second noon was taken for granted as a day. Then humans wanted to know the exact moment of the day, so they invented various timekeeping tools, or clocks. However, these methods were not highly accurate, and it wasn't until the development of electronic oscillators and atomic clocks in the 20th century that timekeeping became much more precise.

Measurement of Time

Time is a phenomenon by which human beings observe and record changes in the environment and the universe.

For example : The time between one sunrise and the next is called a day, a month is measured from one new moon to the next, and a year is fixed as the time taken by the earth to complete one revolution around the sun.

Often we need to measure intervals of time which are much shorter than a day. Clocks or watches are the most common time measuring devices.

One of the earliest means of measurement of time was with the help of movement and change in the position of the sun during the day, and the moon during the night. The position of stars also helped in estimating time during night. Jantar Mantar located in Central Delhi is an advanced form of a Sundial.

The device used to measure time is known as a clock. The older clocks like the sand clock, the water clock, were somewhat crude type of clocks.

They only gave an approximation of a duration of time. Sundials were more accurate than the sandclocks and the water clocks. Nowadays, we use a wrist watch or quartz clock. A mechanical



Sundial